# ECE 310: Digital Signal Processing

Lecture 7

## $z$-transform properties

Corey Snyder

## Learning Objectives

After this lecture, you should be able to:

- Understand key properties of the $z$-transform and the ROC.
- Apply properties of the $z$-transform to derive the $z$-transform for discrete-time signals.

## Recap from previous lecture

We motivated and introduced the $z$-transform in the previous lecture. We saw that the $z$-transform is a powerful tool for determining the response of LTI systems to complex exponential inputs. We will continue in this lecture by further defining the $z$-transform, region of convergence, and their key properties.

## 1 Overview of the $z$-transform

We define the $z$-transform of a discrete time signal $x[n]$ as

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}. \tag{1}$$

This sum is not guaranteed to converge for all values of $z$. Thus, we must specify a region of convergence (ROC) to identify where $X(z)$ is well-defined. Moreover, we saw in the previous lecture that the ROC is necessary to make each $X(z)$ unique. It is important to note that $X(z)$ is not defined for values of $z$ that lie outside the ROC *even if the expression for $X(z)$ computes a finite value.* Consider, for example,

$$x[n] = \left(\frac{1}{2}\right)^n u[n] \overset{\mathcal{Z}}{\Longleftrightarrow} X(z) = \frac{1}{1-\frac{1}{2}z^{-1}},\ |z| > \frac{1}{2}. \tag{2}$$

The above $X(z)$ evaluates to $-1$ for $z = \frac{1}{4}$. However, let's check the original $z$-transform sum for this signal $x[n]$:

$$X\left(\frac{1}{4}\right) = \left(\sum_{n=-\infty}^{\infty} x[n]z^{-n}\right)\Bigg|_{z=\frac{1}{4}} \tag{3}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n \left(\frac{1}{4}\right)^{-n} \tag{4}$$

$$= \sum_{n=0}^{\infty} 2^n. \tag{5}$$

Clearly, the $z$-transform sum does not converge for $z = \frac{1}{4}$, which is consistent with the given ROC.

Let's summarize some key properties that define the $z$-transform and its ROC:

1. Values of $z$ where $X(z) = 0$ are known as zeros while values where $X(z) \to \infty$ are referred to as poles.
2. The ROC defines values of $z$ for which the $z$-transform sum converges. Thus, the ROC cannot include any poles.
3. The ROC is a connected region, i.e. you can connect any two points in the ROC with a (curvy) line without exiting the ROC.
4. For infinite-length signals, the ROC can have one the following shapes:
   - Right-sided sequence, i.e. $x[n] = 0,\ n < n_0$: ROC=$|z| > a$.
   - Left-sided sequence, i.e. $x[n] = 0,\ n > n_0$: ROC=$|z| < a$.
   - Two-sided: ROC=$a < |z| < b$.
5. For finite-length signals, the ROC is the entire complex $z$-plane, except perhaps $z = 0$ for causal sequences or $z = \infty$ for non-causal sequences. For example, the causal sequence $x[n] = \delta[n] + \delta[n-1]$ has $X(z) = 1 + z^{-1}$ where the ROC is $|z| > 0$. Likewise, the non-causal sequence $x[n] = \delta[n+1] + \delta[n]$ has $X(z) = z + 1$ and the ROC of $|z| < \infty$. If we add these two sequences, we would have an ROC of $0 < |z| < \infty$.

Using the above rules, we may concisely describe the ROC for infinite-length signals using their pole locations. For a right-sided signal, we know that its ROC will be the exterior of a circle and its ROC cannot contain any poles. Therefore, the ROC of a right-sided signal starting at $n \geq 0$ must be $|z| > |p_{\max}|$ where $|p_{\max}|$ is the largest magnitude pole of the signal. Conversely for left-sided signals, the ROC must be the interior of a circle and cannot contain any poles. Therefore, the ROC of a left-sided signal ending at $n \leq -1$ must be $|z| < |p_{\min}|$ where $|p_{\min}|$ is the smallest magnitude pole of the signal. We will revisit these rules in greater detail in upcoming lectures, i.e. why do the start/end points matter for right/left-sided signals and two-sided signals, when discussing BIBO stability, but for now we have some quick ways of determining the ROC of a given $z$-transform.

## 2 Key properties of the $z$-transform

We have computed the $z$-transform thus far using the $z$-transform sum or by inspection using a table of transform pairs. This is of course quite limiting. We would like to quickly compute the new $z$-transform of a signal if we scale it, shift it, reverse it, add to it, and so on. In this section, we will list several important $z$-transform properties along with a few proofs while further derivations and examples are reserved for lecture. For the below properties, let $R_x$ denote the ROC for signal $x[n]$.

**Time-shifting.**

$$x[n-k] \overset{\mathcal{Z}}{\Longleftrightarrow} z^{-k}X(z) \tag{6}$$
$$\text{ROC} = R_x \text{ (except possibly } z = 0 \text{ or } z = \infty).$$

To derive the time-shifting property, let $y[n] = x[n-k]$.

$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]z^{-n} \tag{7}$$
$$= \sum_{n=-\infty}^{\infty} x[n-k]z^{-n} \tag{8}$$
$$= \sum_{m=-\infty}^{\infty} x[m]z^{-(m+k)} \tag{9}$$
$$= z^{-k} \sum_{m=-\infty}^{\infty} x[m]z^{-m} \tag{10}$$
$$= z^{-k}X(z). \tag{11}$$

Above, line 9 follows from substituting $n = m + k$.

**Linearity.**

$$ax_1[n] + bx_2[n] \overset{\mathcal{Z}}{\Longleftrightarrow} aX_1(z) + bX_2(z) \tag{12}$$
$$\text{ROC} \supseteq R_{x_1} \cap R_{x_2}.$$

This simply states that the $z$-transform is a linear transform. The superset symbol $\supseteq$ indicates that the ROC contains at least the intersection of each signal's ROCs. This makes intuitive sense since we only need one $z$-transform sum to diverge in order to make the entire $z$-transform diverge. The statement of "at least" refers to exceptions where we have certain pole-zero interactions between the two signals. Most typically, the ROC is simply the intersection of the two ROCs. More on this in later lectures.

**Convolution.**

$$x_1[n] * x_2[n] \overset{\mathcal{Z}}{\Longleftrightarrow} X_1(z)X_2(z) \tag{13}$$
$$\text{ROC} \supseteq R_{x_1} \cap R_{x_2}.$$

To derive the convolution property, let $y[n] = x_1[n] * x_2[n]$.

$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]z^{-n} \tag{14}$$
$$= \sum_{n=-\infty}^{\infty} \left( \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k] \right) z^{-n} \tag{15}$$
$$= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k]z^{-k}z^{-(n-k)} \tag{16}$$
$$= \sum_{m=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_1[k]x_2[m]z^{-k}z^{-m} \tag{17}$$
$$= \sum_{m=-\infty}^{\infty} x_2[m]z^{-m} \sum_{k=-\infty}^{\infty} x_1[k]z^{-k} \tag{18}$$
$$= X_1(z)X_2(z). \tag{19}$$

Above, line 17 follows from the substitution $n = m + k$.

*This is a remarkable property!* We now can say that convolution in the time-domain corresponds to multiplication in the $z$-domain. This property will be key when we further discuss transfer functions and the characterization of LTI systems using the $z$-transform.

**Lecture exercise**: Prove the time shifting property using the convolution property.

**Differentiation.**

$$nx[n] \overset{\mathcal{Z}}{\Longleftrightarrow} -z\frac{dX(z)}{dz} \tag{20}$$
$$\text{ROC} = R_x.$$

The differentiation property will be useful to us when we discuss double-poles and stability of LTI systems in upcoming lectures.

**Conjugation.**

$$x^*[n] \overset{\mathcal{Z}}{\Longleftrightarrow} X^*(z^*) \tag{21}$$
$$\text{ROC} = R_x.$$

Table 1: Useful $z$-transform properties.

| Property | Signal | $z$-transform | ROC |
|---|---|---|---|
| Time-shifting | $x[n-k]$ | $z^{-k}X(z)$ | $R_x$ except $z=0$ or $z=\infty$ |
| Linearity | $ax_1[n]+bx_2[n]$ | $aX_1(z)+bX_2(z)$ | At least $R_{x_1}\cap R_{x_2}$ |
| Convolution | $x_1[n]*x_2[n]$ | $X_1(z)X_2(z)$ | At least $R_{x_1}\cap R_{x_2}$ |
| Differentiation | $nx[n]$ | $-z\frac{dX(z)}{dz}$ | $R_x$ |
| Conjugation | $x^*[n]$ | $X^*(z^*)$ | $R_x$ |
| Time reversal | $x[-n]$ | $X\left(z^{-1}\right)$ | $1/R_x$ |
| Scaling | $a^n x[n]$ | $X\left(z/a\right)$ | $\lvert a\rvert R_x$ |
| Real-part | $\mathrm{Re}\{x[n]\}$ | $(1/2)[X(z)+X^*(z^*)]$ | At least $R_x$ |
| Imaginary-part | $\mathrm{Im}\{x[n]\}$ | $(1/2)[X(z)-X^*(z^*)]$ | At least $R_x$ |

**Time reversal.**

$$x[-n] \overset{\mathcal{Z}}{\Longleftrightarrow} X\left(z^{-1}\right) \tag{22}$$
$$\mathrm{ROC} = \frac{1}{R_x}.$$

**Lecture exercise**: Prove the time reversal property.

For convenience, we summarize the above properties and a couple additional properties in Table 1, which corresponds to Table 3.2 in the *Manolakis and Ingle* textbook.